

Entrepreneurial Finance & Venture Capital

Session 3 · Lecture Notes | VC Valuation, Cap Tables & Staged Financing

1. Session Overview

Session 3 has two parts. The first half reviews the **homework on ASP.com** — a worked example of calculating asset intensity from a real balance sheet. The second half introduces the **VC valuation method**: how investors back out a current valuation from a target exit, and how multiple funding rounds, dilution, and cap tables work in practice.

- Key theme: valuation is a *negotiated outcome*, not a purely mechanical calculation.
- The VC method is 'top-down' — start from the exit, work backwards — versus the DCF's 'bottom-up'.
- Both approaches embed the same underlying logic; they are two presentations of the same idea.

2. Homework Review — ASP.com Asset Intensity Calculation

The homework asked students to calculate the asset intensity (NOA/Sales) of **ASP.com**, a SaaS/subscription software company, using its 2017–2018 balance sheet. The class distribution of answers ranged widely — the two most common mistakes were including cash and omitting deferred revenue.

Step 1 — Identify What to Include

The goal is to capture only **operating** assets and liabilities — items needed to run the business day-to-day. Financing items (debt, equity) and non-operating items (excess cash) are excluded.

Balance Sheet Item	Include?	Reason
Cash & equivalents	■ Exclude	Excess cash — not needed for operations in a software business. Only include if cash is operationally necessary (e.g. a bank teller, foreign exchange desk, or grocery till).
Accounts receivable (A)	■ Include	Cash owed by customers — directly tied to operations.
Prepaid expenses (B)	■ Include	Operational prepayments — part of working capital.
Property & equipment (C)	■ Include	Net fixed assets — needed to run operations.
Term receivables non-current (D)	■ Include	Long-term receivables — a sale has been made and an invoice sent. The accounting classification as 'long-term' is irrelevant to cash-flow analysis.
Accounts payable (E)	■ Include	Operating liability — offsets assets in NOWC.
Commissions payable (F)	■ Include	Operating liability — employee costs accrued.

Line of credit / Debt (G)	■ Exclude	Financing decision — not part of operating cash flow.
Current revenue (G)	deferred ■ Include	Cash received from prepaid subscriptions — a key operating liability that lowers NOWC.
Long-term revenue (H)	deferred ■ Include	Same as above but for multi-year subscriptions. Still cash in hand from operations — the long-term classification is an accounting convention, not an economic one.

Step 2 — Calculate NOWC, NFA, NOoLTA

Using 2018 figures (\$thousands):

NOWC = (A+B) – (E+F+G)	$(14,439 + 509) - (1,025 + 1,636 + 26,127) = 14,948 - 28,788 = -13,840$
NFA = C	Property & equipment, net = 2,942
NOoLTA = D – H	Term receivables – Long-term deferred revenue = $2,278 - 9,426 = -7,148$
NOA = NOWC + NFA + NOoLTA	$-13,840 + 2,942 + (-7,148) = -18,046$
Sales (2018)	35,879
Asset Intensity = NOA / Sales	$-18,046 / 35,879 = -50.3\%$

***Why is it negative?** Customers pay upfront (deferred revenue) before ASP.com delivers the service. This means customer cash finances the business rather than the business having to tie up its own cash. The larger the deferred revenue, the more negative NOWC becomes — and the more cash the business generates as it grows. This is the same pattern we saw with Strava.*

Common Mistakes

- **Including cash:** adds to assets → artificially inflates asset intensity.
- **Excluding deferred revenue:** removes a key liability → reduces operating liabilities → inflates NOWC → makes the company look more asset-intensive than it is.
- Both errors together can shift the answer from –50% to +50% — a 100 percentage-point swing.

***Prof. Nanda's analogy:** cash on the balance sheet is like fat on a body — it is excess energy stored, not the calories needed to keep the engine running. Operating cash flow needs are the calories required to stay alive. We measure operating needs only; we don't count stored fat.*

3. Applying the 3-Stage Model — RightNow Technologies

The homework also asked students to run the 3-stage FCF model using ASP.com's data as the starting point (Year 0). The model is based on a real company: **RightNow Technologies**, a SaaS CRM business acquired by Oracle in Year 8 for **\$1.5 billion**.

Parameter	Stage 1 (Yrs 1–5)	Stage 2 (Yrs 6–10)	Terminal (Yr 11+)
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Growth (g)	50%/yr	25%/yr	3%/yr
EBIAT / Sales (p)	2.4%	3.5%	5.0%
NOA / Sales (a)	−50%	−50%	−50%

Starting conditions (Year 0): Sales = \$35.9M, EBIAT/S = −9.4%, NOA/S = −50.3%

Key Outputs from the Model

- By Year 10: Sales grow to ~\$927M — roughly 26× the starting figure.
- FCF is **strongly positive throughout** because asset intensity is −50%: growing faster actually generates more cash (the subscription paradox from Session 2).
- Enterprise Value at Year 0: ~\$637M at a 10% WACC.
- Revenue multiple (EV/Sales) starts at ~10.6× and converges to ~0.9× as growth slows — illustrating why high-growth companies command premium revenue multiples.

Real-world check: Oracle acquired RightNow in Year 8 for \$1.5B, with asset intensity still at −40%. The model's aggressive growth projections actually underestimated the outcome — a reminder that even with bold assumptions, the best companies can still surprise to the upside.

Why Asset Intensity Might Change Over Time

- If the customer mix shifts toward large Fortune 500 clients, those customers have more negotiating power and may pay less upfront or demand longer credit terms.
- This would reduce deferred revenue → NOWC becomes less negative → asset intensity rises.
- Conversely, if the subscription model is preserved and CapEx stays low, highly negative asset intensity is sustainable long-term.

Always link your asset intensity assumption to a specific view of the business model — it cannot be set in isolation.

4. Industry Comparison — p, a, g Across Five Companies

The lecture walks through five real companies (FY2013–14) to show how dramatically p, a, and g vary by industry — and how that drives valuation multiples.

Company		EBIAT/Sales (p)	Asset Intensity (a)	Growth (g)	Mkt Value/Sales	Key Insight
DAL Delta Airlines		6–8%	+28%	7%	0.9×	Huge fixed assets (planes) but negative NOWC (tickets pre-paid). Asset-heavy overall.
BBY Best Buy		2–2.4%	+5%	−1%	0.3×	Razor-thin retail margins, flat growth. Low valuation multiple.

MSFT Microsoft	19–22%	–2.5%	12%	4.4×	Software subscription model → negative asset intensity. High margins + growth = premium multiple.
AMGN Amgen	21–22%	+24–34%	7%	6.7×	High margins (patent-protected drugs) but asset-intensive R&D/manufacturing. Valued on earnings quality not asset efficiency.
PG Procter & Gamble	12–13%	+30%	1%	2.5×	Steady consumer staples business. Moderate margins, positive asset intensity, low growth.

Key lesson from the comparison: Microsoft's –2.5% asset intensity combined with 12% growth produces the highest revenue multiple (4.4×) despite not having the highest margins. Asset intensity and growth interact — for negative-a businesses, more growth = more cash, which is why the market rewards them so highly.

- Airlines pre-sell tickets → negative NOWC, but huge physical assets (planes) dominate.
- Southwest vs Delta: Southwest's faster aircraft turnarounds improve asset utilisation, lowering asset intensity and allowing lower prices — a direct p-a-g trade-off.
- Biotech (Amgen): high margins from patent protection, but asset-intensive manufacturing — the market values the earnings stream, not the balance sheet efficiency.

5. Why DCF Fails for Startups — Context for the VC Method

Before covering the VC method, the lecture sets the stage for why standard tools break down.

No historical data	Startups may have zero revenue. DCF requires projecting from a baseline that doesn't exist.
Negative or volatile profits	The FCF formula starts with EBIAT. If EBIT is deeply negative, the starting point is meaningless.
High uncertainty / power law	72% of VC-backed companies return less than invested. A few (5%) generate >10× — standard sensitivity analysis around one scenario misses the distribution entirely.
No observable beta	Private companies have no share price, so systematic risk cannot be measured. Convention: use ~10% WACC for the 'success scenario'.

Explosive early growth	A 50%/yr growth rate in a terminal value formula produces infinite value. The 3-stage model handles this by letting g converge to 3% only in the terminal stage.
Valuation = negotiation	The agreed price reflects fundamentals <i>and</i> relative bargaining power. A VC with a hot fund has more leverage; a founder with competing term sheets has more.

Power law context: 72% of investments return less than the capital invested. 12% return between 1–3×. Only 5% return more than 10×. Yet that 5% generates over 50% of all gross returns. The VC model is designed around this — you need one massive win to cover all the losses.

6. VC Fund Structure & Compensation

Fund structure	LPs (pension funds, endowments, family offices) contribute ~99% of capital. The VC firm acts as General Partner (GP), managing the fund.
Fund life	Typically 10 years with extension clauses. Capital is locked up — illiquid.
Management fee	2% of committed capital per year (falling to ~1% in later years). Covers salaries and operations regardless of performance.
Carried interest	20% of profits above a hurdle return. The GP's real incentive — aligns with fund performance.
'2 and 20'	The shorthand for management fee + carry. Industry standard.
Call option incentive	Carry is like a call option — the GP captures upside but doesn't share downside beyond reputation. This drives preference for risky, high-upside bets: a 1% chance of £1bn is preferred over £10M for certain.

Why this matters for founders: VC incentive structures mean they need large exits — a \$50M outcome may be fine for the founder but useless for a \$500M fund. This drives VCs to push for faster growth and bigger swings, even when a smaller outcome might be optimal for the entrepreneur.

7. The VC Valuation Method

Core Logic

The VC method works **backwards from a target exit value**. Rather than projecting cash flows forward, you ask: 'What must I own at exit to hit my required return, given what I invest today?'

Step 1 — Target exit value	Estimate revenues at exit (e.g. Year 5). Apply a comparable revenue multiple (e.g. 6× revenue = \$150M exit value).
Step 2 — Required future value	Compound the investment at the target IRR: $\text{Investment} \times (1 + \text{IRR})^n$

Step 3 — Required ownership at exit	Required Future Value / Exit Value. This is the % the VC must own at exit to hit their return.
Step 4 — Post-money valuation	Investment / Required Ownership %. This is what the company is worth today — implied by the deal terms.

Worked Example — Single Round

Investment = \$4M, IRR target = 50%, horizon = 5 years, exit = 6× \$25M revenue = \$150M:

Required future value = $4 \times (1.5)^5 = \$30.4\text{M}$

Required ownership at exit = $30.4 / 150 = 20\%$

Post-money valuation = $4\text{M} / 20\% = \$20\text{M}$

Effect of IRR on Valuation

IRR Target	Required Future Value	Ownership Needed	Post-Money Valuation	Implication
25%	$4 \times (1.25)^5 = \$12.2\text{M}$	8.1%	\$49M	Cheap money → high valuation → founder retains more
50%	$4 \times (1.50)^5 = \$30.4\text{M}$	20%	\$20M	Standard VC terms
75%	$4 \times (1.75)^5 = \$66\text{M}$	44%	\$9M	Expensive money → low valuation → founder heavily diluted

Key insight: the IRR target is a proxy for perceived risk. Earlier-stage investments attract higher IRR targets (75%+) because the probability of failure is higher. Later-stage investments use lower rates (25–40%) because more proof points exist. Higher risk → lower valuation → more dilution for the founder.

Where Do Valuation Increases Between Rounds Come From?

Each funding round should reflect a step-up in valuation (an 'up round'). This happens when the company hits proof points that reduce the probability of failure:

- Product milestones reached (MVP shipped, beta users acquired)
- First paying customers or signed contracts
- Customer buzz, press, or endorsements from credible sources
- Anything that makes investors more confident the company won't go to zero

Example: if success value = \$200M and p(success) rises from 5% to 20%:

Value at 5% = $0.05 \times 200 = \$10\text{M}$ → **Value at 20%** = $0.20 \times 200 = \$40\text{M}$

The \$30M of new value was created by de-risking, not by growing revenues.

8. Multiple Funding Rounds & Dilution

When multiple rounds of investment are anticipated, each early investor must take a **larger stake today** to account for the dilution they will suffer in future rounds.

Worked Example — Three Rounds

VC1 invests \$1.5M (IRR = 50%), expecting VC2 to invest \$1M in Year 2 (IRR = 40%) and VC3 to invest \$1M in Year 4 (IRR = 25%). Exit: \$7.5M revenue $\times 5\times$ = \$37.5M in Year 5.

Investor	IRR	If Only Investor	Diluted By	Adjusted Stake
VC3 (Round 3)	25%	3.3%	Nothing	3.3%
VC2 (Round 2)	40%	7.3%	3.3% (VC3)	$7.3\% / 96.7\% = 7.6\%$
VC1 (Round 1)	50%	30.4%	$3.3\% + 7.3\% = 10.7\% \rightarrow$ retains 89.3%	$30.4\% / 89.3\% = 34\%$

Calculating New Shares

Starting with 1,000,000 founder shares:

New shares = [% ownership / (1 – % ownership)] $\times$ existing shares

Round 1 (VC1, 34%)	$34\% / 66\% \times 1,000,000 = 515,055$ new shares @ \$2.91/share
Round 2 (VC2, 7.6%)	$7.6\% / 92.4\% \times 1,515,055 = 124,077$ new shares @ \$8.06/share
Round 3 (VC3, 3.3%)	$3.3\% / 96.7\% \times 1,639,132 = 56,522$ new shares @ \$17.69/share

9. Capitalisation Table (Cap Table)

A cap table is the definitive record of who owns what at each stage of the company's life. It tracks shares, ownership %, cash invested, and implied valuations across all rounds.

	Founders	Round 1	Round 2	Round 3
Entrepreneur	1,000,000 (100%)	1,000,000 (66%)	1,000,000 (61%)	1,000,000 (59%)
Investor 1	—	515,055 (34%)	515,055 (31.4%)	515,055 (30.4%)
Investor 2	—	—	124,077 (7.6%)	124,077 (7.3%)
Investor 3	—	—	—	56,522 (3.3%)
Total shares	1,000,000	1,515,055	1,639,132	1,695,654
Pre-money valuation	\$1M	\$2.9M	\$12.2M	\$29.0M
Cash in	—	\$1.5M	\$1.0M	\$1.0M
Post-money valuation	\$1M	\$4.4M	\$13.2M	\$30.0M
Price per share	—	\$2.91	\$8.06	\$17.69

Note on share price: share price only has meaning within a company. Comparing \$17.69 here to \$17.69 in another company is meaningless — what matters is the post-money valuation and your % ownership. Post-money = latest valuation mark, but it's illiquid paper money, not cash.

10. Two Approaches to Discount Rates

Approach A High discount rate on best-case scenario

Model the success path (e.g. using the 3-stage model). Apply a high discount rate (50–75%) to reflect the probability of failure. Fast and practical — the heuristic used by most practitioners. *Risk*: the high rate is opaque — it's hard to know what probability of success is implied.

Approach B Scenario-weighted DCF

Model 2–3 scenarios (success, moderate, failure). Assign probabilities to each. Discount at a reasonable rate (~10% WACC). More transparent — you can see exactly what probability of success drives the valuation. *Risk*: harder and more time-consuming to build.

They are equivalent in logic. A 50% discount rate on the success scenario implicitly assumes a certain probability of success. You can back out that implied probability by asking: 'At what probability does the probability-weighted NPV equal the high-rate DCF value?' Going back and forth between the two approaches is a useful sanity check.

- Example: 3-stage model gives \$637M in the success scenario at 10% WACC. If a VC applies 50% discount rate, the implied probability of reaching that path can be calculated. That number should pass a gut-check — does it seem like a reasonable estimate of success probability?
- Applying 50% discount rate to Stages 1–5 is reasonable. But applying it to Stage 3 (terminal, Year 11+) is wrong — by then the company is mature and a 10% WACC is appropriate.

11. Key Takeaways

Asset intensity calculation

Exclude cash (unless operationally required) and include ALL deferred revenue (short AND long-term). Getting these two wrong can shift the answer by 100 percentage points.

NOA = NOWC + NFA + NOoLTA

Asset Intensity = NOA / Sales. For ASP.com: –50.3%, driven by large deferred revenue liability.

VC method

Work backwards from exit: $\text{Investment} \times (1 + \text{IRR})^n / \text{Exit Value} = \% \text{ ownership needed}$. Post-money = Investment / % ownership.

IRR as risk proxy

Higher IRR targets → lower valuation → more dilution. Early stage = 75%+; later stage = 25–40%. IRR captures execution risk, not just time value.

Dilution anticipation

Early investors must take larger stakes today to compensate for future dilution. Formula: $\text{adjusted stake} = \text{base stake} / (1 - \text{future dilution } \%)$.

Cap table

Tracks shares, % ownership, and valuations across all rounds. Share price is meaningless across companies — post-money valuation is what matters.

Two discount approaches

High rate on one scenario $\equiv$ probability-weighted scenarios at WACC. Same underlying logic, different presentation. Use both as a cross-check.

Valuation is negotiated

Fundamentals set a range; bargaining power determines where in that range the deal lands. A VC with a hot fund has more leverage. A founder with competing bids has more.

All three sessions covered — POCD & Strava (S1) · FCF drivers & 3-stage model (S2) · VC valuation, cap tables & staged financing (S3).